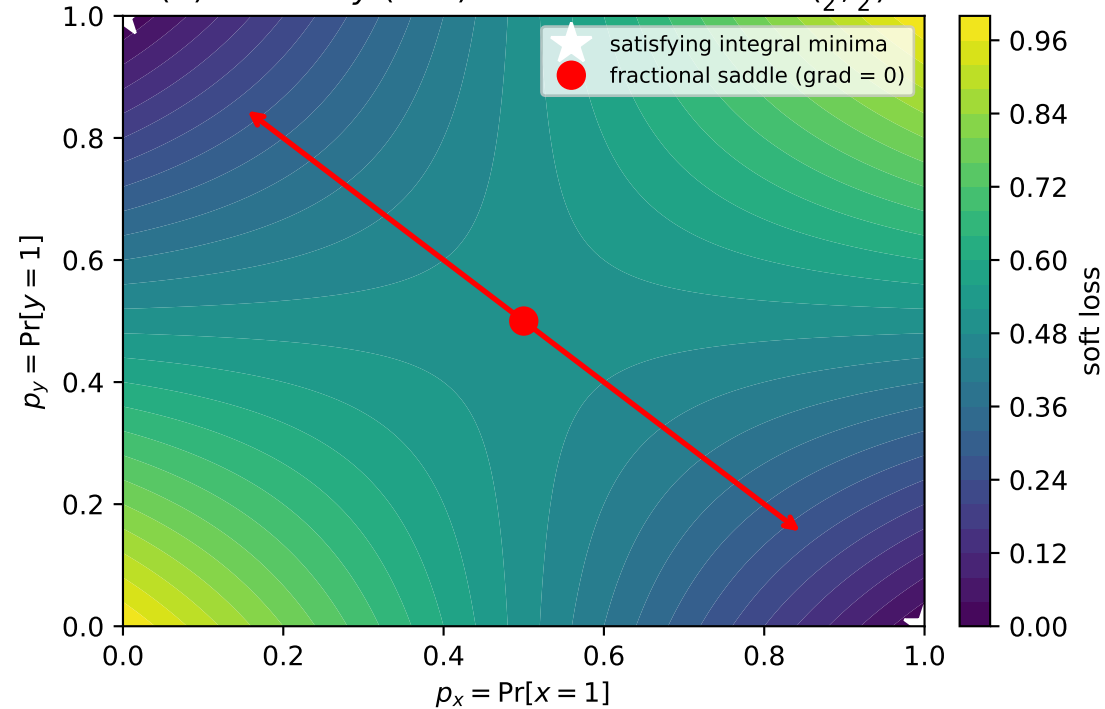


Fractional local minima: why first-order relaxation stalls and region materialization is needed

(A) XOR $x \neq y$ (SAT): fractional saddle at $(\frac{1}{2}, \frac{1}{2})$



(B) Temperature-regularized free energy F_τ (XOR): minimizer is fractional until $\tau \rightarrow 0$

